MIL-I-18079B(AS)
31 March 1969
SUPERSEDING
MIL-I-18079A(Aer)
19 March 1958

MILITARY SPECIFICATION

INSTALLATION OF ANGLE OF ATTACK AND SIDESLIP SYSTEMS

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - This specification covers general requirements for the installation of angle of attack and sideslip systems which involve the use of transmitters of local angle of attack or local angle of sideslip.

* 1.2 Classification - The angle of attack and angle of attack and sideslip systems shall be of the following types, as specified:

Type I - Without compensation for position errors.
Type II - With local angle of sideslip

Type III - With Approach Power Control

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-7080	Electric Equipment, Aircraft, Selection and Installation of
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

FSC 6610

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MIL-I-18856	Indicator, Angle of Attack, 28 Volt D. C.
MIL-T-19229	Transmitter, Angle of Attack or Sideslip, Local
MIL-C-23866	Control Set; Approach Power AN/ASN-54(V)

STANDARDS

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-203	Cockpit Controls, Location and Actuation of for Fixed Wing Aircraft
MIL-STD-411	Aircrew Station Signals
MS25317	Indexer, Approach
MS25318	Light, Approach, 28V
MS28062	Indicator, Angle of Attack, 2-3/4 Dial, 28 Volt D. C.
MS28066	Transmitter, Angle of Attack or Sideslip, Local
MS28067	Indicator, Angle of Attack, 1-7/8 Dial, 28 Volt D. C.
MS33586	Metals, Definition of Dissimilar
AN507	Screw, Machine, Flat Head, 100 Degree
AN6230	Gasket, O Ring Hydraulic

2.2 Availability of Documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Materials - Materials used in the installation of angle of attack and angle of attack and sideslip systems shall be of high quality, suitable for the prupose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided they are approved by the procuring activity and contain provisions for adequate tests. The use of contractor's specifications will not constitute waiver of Government inspection.

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3.1.1 Metals - All metals used in the installation of angle of attack and angle of attack and sideslip systems shall be suitably protected to resist corrosion during normal service life.

* 3.1.1.1 Dissimilar Metals - Where practicable, dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other. Where such contacts are unavoidable, both contact surfaces shall be painted with Zinc Chromate Primer conforming to MIL-P-8585. The installation shall be made with cadmium-plated mounting screws while the primer is still wet. Care shall be exercised to keep primer off the electrical connections.

* 3.1.2 Selection of Materials - Specifications and standards for all materials, parts and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in 3.1.2.1.

3.1.2.1 Standard Parts - Standard parts (MS or AN) shall be used wherever they are suitable for the purpose and shall be identified on any drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration. The corresponding standard part numbers shall be referenced in the parts list and, if practicable, on contractor's drawings. If there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all the requirements of this specification.

3.1.3 Wiring - All electrical wiring shall be in accordance with MIL-W-5088, and Figure 1.

3.2 Types of Systems - An angle of attack or an angle of attack and sideslip system, may consist of one of the combinations of components shown in Table I.

3.3 System Installation - All components of an angle of attack or an angle of attack and sideslip system shall be installed in accordance with MIL-E-7080 and in accordance with the requirements for the type system as specified herein.

3.3.1 Transmitter Installations -

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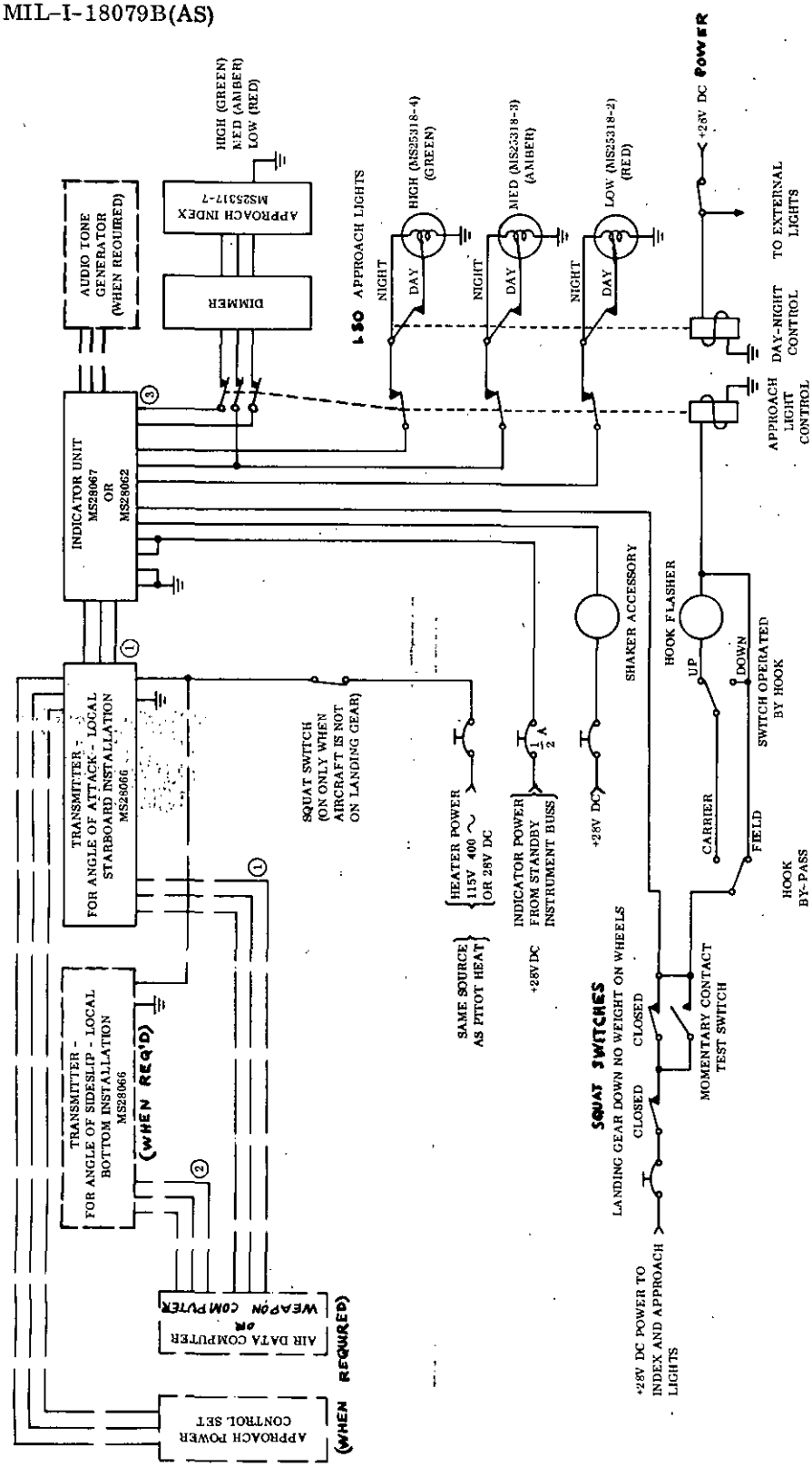


Figure 1. Wiring - Elementary Installation

NOTES:
1. + AND - LEADS TO TRANSMITTER POTENTIOMETER ARE REVERSED BETWEEN PORT AND STARBOARD INSTALLATIONS.
2. + AND - LEADS TO TRANSMITTER POTENTIOMETER ARE REVERSED BETWEEN TOP AND BOTTOMSIDE INSTALLATIONS.
3. THE GRADUATIONS ON THE INDICATOR REFLECT THE ANGLE OF ATTACK IN RELATIVE UNITS, NOT IN DEGREES.

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TABLE I

COMPONENTS OF SYSTEMS

TYPE	I	II	III
Transmitter, Local Angle of Attack, MIL-T-19229	X	X	X
Indicator, Angle of Attack, MIL-I-18856	X	X	X
Approach Index, MS25317-6 or -7	X	X	X
Approach Lights, MS25318-2, -3, and -4	X	X	X
Transmitter, Local Angle of Sideslip, MIL-T-19229 (When required)		X	X
Approach Power Control Set, MIL-C-23866 (When required)			X
Dimming Rheostat (Approach Index)	X	X	X
Accessory, Stall Warning	X	X	X

3.3.1.1 Location - Any transmitter for local angle of attack or sideslip shall be located in a position on the aircraft which is free of turbulent air for all operational combinations of angle of attack, sideslip and Mach number for the aircraft. The transmitter shall be so located and aligned that the direction of local airflow in all flight conditions is included within the range of operation of the transmitter. Insofar as practicable and consistent with the above requirements, the location of a transmitter should minimize the effects of aerodynamic interference and exposure to spray, dirt, gun or rocket fire, and accidental damage by aircraft handling personnel and crew. For angle of attack transmitters, locations least affected by sideslip are preferred and in no case shall the effect of sideslip exceed the tolerance established in 4.3.5. For sideslip transmitters, the sensing element shall be located in the plane of symmetry of the aircraft. In the systems, consideration of the effects of aerodynamic interference upon the dependence of position error on Mach number should influence the selection of transmitter locations. In general, the largest effect of this kind is exerted by the aircraft wing at the position of the local angle of attack transmitter and the latter should be located as far forward of the wing as other requirements and considerations permit.

3.3.1.2 Mounting - The method of mounting shall include provisions equivalent to those shown in Figure 2 for external installation of the transmitter, vernier adjustment of predrilled dowel locating holes and adequate provision for pressure sealing up to 15 psi.

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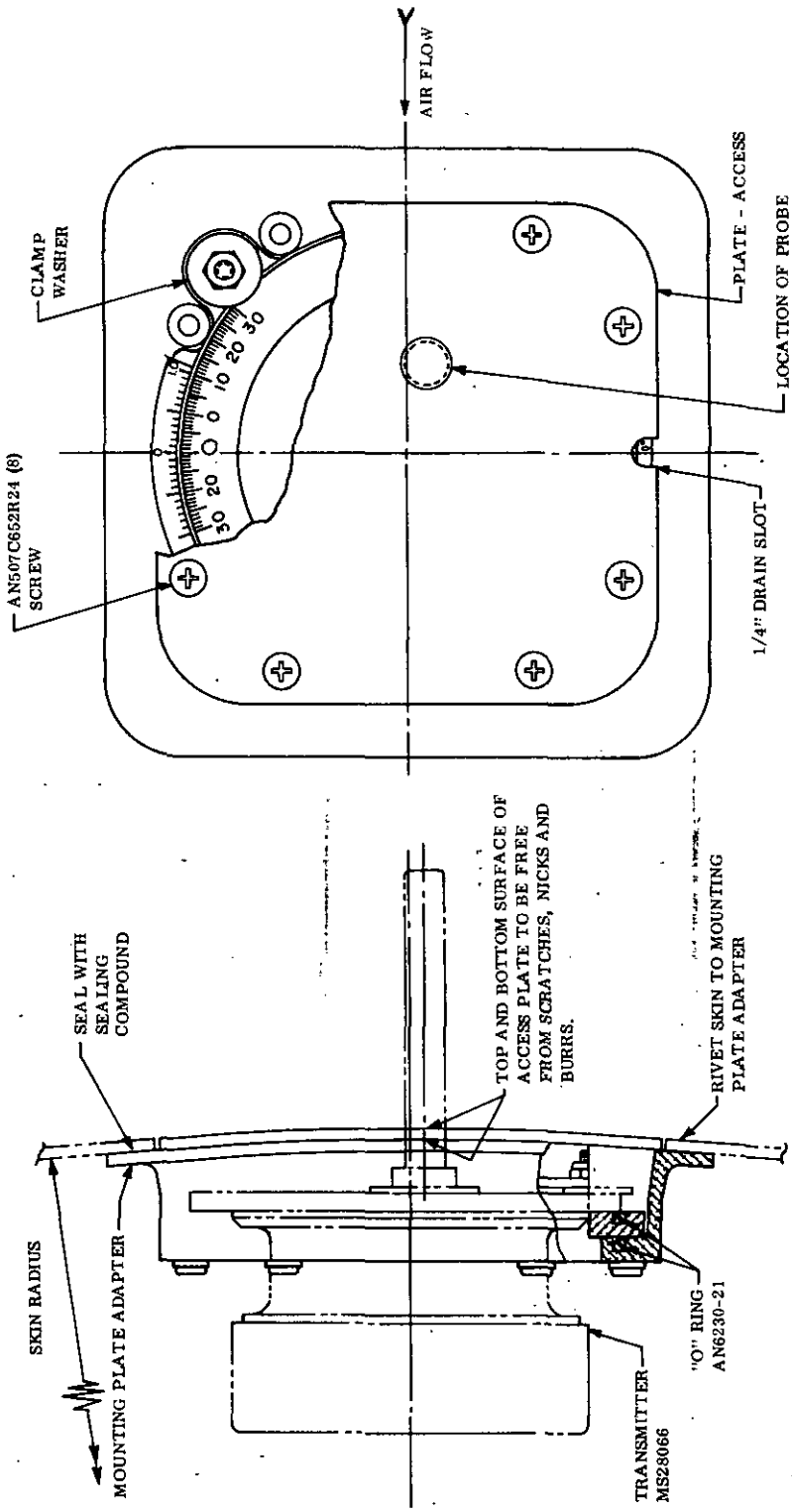


Figure 2. Transmitter Installation (typical)

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3.3.1.3 Dowel Alignment - For a local angle of sideslip transmitter, the centers of the dowel holes in the mounting plate adapter shall be on a transverse line which makes an angle of less than one milliradian with a perpendicular to the aircraft plane of symmetry. For a local angle of attack transmitter, the transverse line through the dowel holes in the mounting plate adapter should be aligned and locked angularly as determined in 4.3.3.

3.3.1.4 Boresight Sideslip Installation - All boresighting may be done on the deck. After the mounting plate adapter has been installed and centered on the plane of symmetry, align the aircraft according to the manufacturers' recommended practice for armament boresighting or position the aircraft to face a target and drop plumb lines at extreme points along the plane of symmetry fore and aft. Position a surveyor's transit between the nose of the aircraft and the target, aligned with the fore and aft plumb lines. Dump the transit telescope and mark the target with a vertical centerline. Install the telescope assembly on the boresight fixture. Set and lock the fixture at 0, then install the boresight fixture in the mounting plate adapter. After loosening the adapter ring which contains the dowel holes, rotate the boresight fixture and adapter ring to align the cross hair in the telescope with the vertical mark on the target. Lock the adapter ring at this point. The transmitter may now be inserted. No further calibration is necessary.

3.3.1.5 Aerodynamic Alignment, Angle of Attack Installation - After the mounting plate adapter has been physically secured to the aircraft, install the transmitter so that the range of angle of attack of the aircraft does not exceed the limits on either end of the particular transmitter used.

* 3.3.1.6 Angle of Attack Transmitter Heater - The angle of attack transmitter heater shall be energized through the landing gear strut compress (squat) switch only when the aircraft is airborne. A by-pass switch at the wheel well shall be used for ground testing of the heater.

Note: This heater circuit can be a part of the pitot heater circuit.

* 3.3.1.7 Additional Signals - When required, the transmitter shall be capable of providing an electrical signal of local angle of attack **or sideslip to other equipment.**

3.3.1.8 Marking - The transmitter locations shall be marked as shown in Figure 3.

3.3.2 Indicator Installation -

3.3.2.1 Location - The angle of attack indicator shall be mounted in the upper left hand corner of the pilot's instrument panel or as otherwise specified by the procuring activity.

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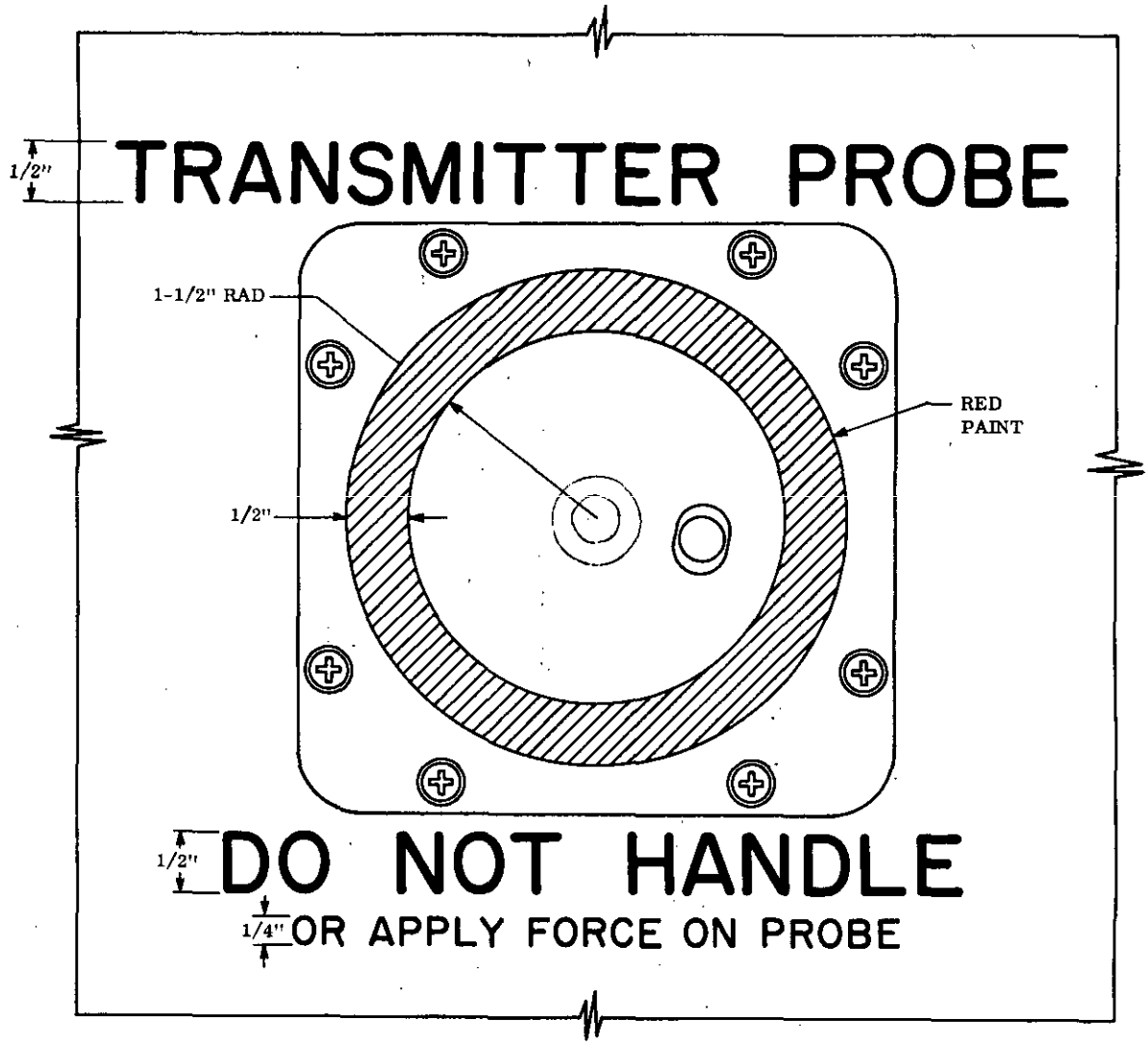


Figure 3. Marking of Transmitter Location

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3.3.2.2 Indicator Dial Adjustment - After fulfilling the requirements of 4.3.3, the indicated angle of attack for power-on landing configuration as specified by the procuring activity shall be set to coincide with the approach reference index by rotating the dial as per applicable instructions.

3.3.2.3 Indicator Switching Adjustment - The switches contained within the indicator shall be set so that the approach range (range when the amber approach light and the center light of the approach indexer are on) shall be 2 indicator dial divisions wide and at the percentage of the stall angle specified by the procuring activity. In addition, the switches for the overlap feature of the approach indexer shall be set so that the high and low light shall be on 1/2 a dial division each during the time the center light is on. The accessory switch shall be adjusted to close at the same dial division as shall be specified in 3.3.6.1.

3.3.2.4 Indicator Reference Indices Settings - The three reference indices (triangular, square, and hatched triangle) shall be set according to the requirements of the procuring activity and flight characteristics of the individual aircraft. The setting of the indices shall be accomplished by removal of the bezel assembly conforming to MIL-I-18856.

3.3.2.5 Multiple Indicating Installations - Where more than one indicator or approach index is warranted, all lights shall be switched by one indicator, which shall act as a master for this purpose.

3.3.3 Approach Indexer Installation -

3.3.3.1 Location - The approach indexer unit shall be mounted atop the glare shield to the left in such a position that it is in the line-of-sight of the pilot and copilot during normal approach conditions. It shall be readily accessible for purposes of relamping.

3.3.3.2 Glare Prevention - Approach indexer installations shall require the use of a hood, where necessary, to prevent glare on the canopy during night maneuvers and to facilitate optimum visibility during day operations.

3.3.3.3 Approach Indexer Dimming - Provision shall be made to dim the approach indexer independently from the other cockpit lights. Further, each lamp in the approach indexer shall have individual dimming. The dimming rheostat shall be mounted on the console with the lighting controls as specified in MIL-STD-203.

* 3.3.3.4 Landing Gear Squat Switch - The indexer lights and the external approach lights are to operate automatically when the landing gear is down and locked with the arresting hook extended. All approach lights and indexer lights are extinguished upon landing by means of a landing gear strut compress (squat) switch. A momentary contact toggle switch with guard labelled "HOOK BY-PASS" is used to by-pass the arresting hook flasher circuit of the approach light system during field mirror landing carrier practice.

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3.3.3.5 Pilot's Test Feature - Provision shall be made for pilot's testing of the approach indexer by means of a press-to-test switch as specified in MIL-STD-411.

3.3.4 Approach Lights shall be installed as specified by the procuring activity or in such a manner that they are visible to the landing signal officer when he is located forward of the beam of the aircraft.

3.3.4.1 Brightness Control - Provision shall be made to control the brightness of the approach lights in two positions for day and night operations.

3.3.4.2 Hook By-Pass Switch - A switch shall be supplied in the cockpit which shall have two positions, viz. CARRIER and FIELD. When set in the CARRIER position, the approach lights will flash if the hook is not down when wheels are down. When set in the FIELD position, the approach lights will remain steady when the wheels are down regardless of the hook position.

* 3.3.5 Approach Power Control Set Installation (when required) shall be installed in accordance with MIL-C-23866.

3.3.6 Accessory, Stall Warning Installation, such as a stick or pedal shaker shall be mounted so that a vibration signal from it can be transmitted to the pilot and copilot if the aircraft should be flown within some per cent of stall as determined by flight test and approved by the procuring activity and without reference to any visual indication within the cockpit.

* 3.3.7 Audio Tone Generator - For the use of a nonverbal auditory angle of attack/airspeed signal, see MIL-STD-411.

* 3.3.8 Computer Installation requiring local angle of attack or sideslip signals shall be electrically connected to the transmitter capable of providing the signal.

3.4 Performance - The installation of an angle of attack or sideslip system shall satisfy the performance requirements specified under paragraph headed Test Methods in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 The angle of attack or sideslip system installation in the first of each model of aircraft submitted for acceptance under contract shall be subjected to all the tests described in this specification which are applicable to the type and class of system installed, prior to installation in the remaining aircraft of the contract, to determine suitability of design and compliance with performance requirements of this specification.

4.2 Test Procedures shall be prepared and submitted to the procuring activity for review and approval.

4.2.1 Individual Tests - Each system installation submitted for acceptance under contract shall be subjected to the tests described herein. In addition, each installation shall be subjected to any other tests specified herein which the Inspector considers necessary to determine conformance with

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the requirements of this specification as installed in the aircraft.

(1) Examination of Installation
(2) System Operation
(3) Indicated Angle of Attack at High Speed
(4) Stall Angle Repeatability
(5) Sideslip Effect
(6) Indicator Switch Settings
(7) Angle of Attack and Sideslip Position Error Calibration
(8) Approach Power Compatibility

4.2.2 Special Tests - The first system installation on an aircraft model and the first installation embodying a change in design, or type, of installation or system and the first installation following a change in the aerodynamic configuration of an aircraft model shall be subjected to all the tests herein described which are applicable to the type of installation.

4.3 Test Methods

4.3.1 Examination of Installation - Each system installation shall be carefully examined to determine conformance with the requirements of this specification not covered by applicable tests.

4.3.2 System Operation - Energize the system and rotate the transmitter slowly to check the performance of the indicator and the audio tone generator, and the signals to the computers and the approach power control. Check the operation of all lights, accessories, controls, switches and heaters. The indicator shall read 30 units when the orifice of the transmitter or the vane transducer is turned to its stop beyond the stall position.

4.3.3 Indicated Angle of Attack at High Speed - Below 12,000 feet altitude in smooth air and level flight at a selected gross weight and indicated airspeed as near V max. for level flight as practicable but not exceeding Mach 0.75, the indicated angle of attack and airspeed shall be noted. This value of indicated angle of attack, after correction for any difference between recorded indicated airspeed and calculated gross weight and the selected values, shall not differ from the nominal value specified for the model of aircraft by more than $\frac{1}{4}$ unit of indicated angle of attack as read by the pilot on the indicator for that model of aircraft.

4.3.4 Stall Angle Repeatability - In the power-on landing configuration, the aircraft shall be brought gradually to a stall with zero sideslip and the angle of attack indicator reading noted at the instant of stall. This procedure shall be repeated. Variation in the pointer position at stall shall not exceed $\frac{1}{2}$ unit.

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4.3.5 Sideslip Effect - The aircraft shall be flown in power-on landing approach condition and configuration and the angle of attack indicator reading noted. Immediately thereafter, the aircraft shall be sideslipped with wings level approximately 5° to the right and the indicator reading noted. The procedure shall be repeated except that the sideslipping shall be to the left instead of the right. The indicator reading during sideslipping shall not differ from that obtained with zero sideslip by more than 3-1/3% of the indicated angle at stall for this configuration per 5 degrees of sideslip.

4.3.6 Indicator Switch Settings - The angle of attack transmitter probe shall be rotated very gradually by hand in the direction of increasing angle of attack. When the approach lights switch from red to amber, the indicated angle of attack shall be noted. Rotation of the probe shall be continued and the indicated angle of attack at which the approach lights switch from amber to green shall be noted. The indicated angle of attack at which the switching occurs shall be in conformance with the requirement of 3.3.2.3. The approach index shall be observed and shall change position in conformance with the requirement of 3.3.2.3.

* 4.3.7 Angle of Attack and Sideslip Position Error Calibration - The complete calibration of position error requires the determination of the relation between the output of an installed local angle of attack transmitter and the true angle of attack of the aircraft and of the relation between the output of an installed local angle of sideslip transmitter and the true angle of sideslip of the aircraft throughout the range of Mach number specified by the procuring activity for the aircraft.

4.3.7.1 Outline of Method - An outline of the method of position error calibration to be used shall be submitted to the procuring activity and approval obtained prior to performing the actual calibration.

4.3.7.2 Accuracy of Position Error Calibration - The position errors of the selected locations for the local angle of attack and sideslip transmitters are considered fully determined when the Mach number dependent functions f_1, f_2 and f_3 in the following formulas have been determined in the range of Mach number specified for the model of aircraft by the procuring activity.

$$\alpha_T = f_1 \alpha_L + f_2$$

$$\beta_T = f_3 \beta_L$$

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where α_L, β_L and α_T, β_T are, respectively, local and true values of angle of attack and sideslip. Unless otherwise specified by the procuring activity, the functions f_1, f_2 and f_3 shall be determined with such accuracy that a variation of any one of these functions from any of its nominal values by its standard deviation (as calculated from the flight test calibration data) shall not change the values of α_T or β_T given by the formulas by more than 2 milliradians when α_L is assigned the value corresponding to buffet onset at the Mach Number and β_L is assigned the value 5 degrees or β_L max for the Mach number, whichever is less.

4.3.8 Approach Power Control Compatibility - *Installations of Type III* systems shall be tested in flight for compliance with the requirements of 3.3.5.

4.4 Rejecton and Retest - The individual installations failing to meet their respective tests shall be rejected. *Installations which have been rejected* may be reworked to correct defects and resubmitted for acceptance. Before the resubmission, full particulars concerning previous rejections shall be furnished the Inspector.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended Use - An installation of an angle of attack system or an angle of attack and sideslip system is intended for use on an aircraft to provide local angle of attack or local angle of sideslip and true angle of attack and true angle of sideslip.

6.2 Changes to Previous Issues - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence applies.

(a) The Contract shall have precedence over any specification.

(b) *This Specification* shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(c) Referenced Specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Project 6610-N011

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-I-18079B(AS) INSTALLATION OF ANGLE OF ATTACK AND SIDESLIP SYSTEMS

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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